Berger Scientific Connectome

The connectome, the book, the method, the session, the origin - in that order.

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color	element	base	where it appears in this paper
red	hydrogen · H 1	adenine	no role in this paper
blue	carbon · C 6	cytosine	no role in this paper
green	nitrogen · N 7	guanine	no role in this paper
orange	oxygen · O 8	hypoxanthine · inosine	oxygen, in the worked example
purple	phosphorus · P 15	uracil	no role in this paper
yellow	sulfur · S 16	thymine	sulfur and cysteine, in the worked example and the ladder
indigo	selenium · Se 34	xanthine · orotate	no role in this paper

Color is a pale field behind the word, never colored type, and it names an element and nothing else.

Abstract. The Berger Scientific Companion is a scientific reference book of 3,543 pages, built to be queried rather than read. It holds 2,680 numbered sections and 143,720 internal links. This manual sets out how to work it: how to frame a question the book can close on, how to run the Berger octave loop of who, what, where, when, how and why, how the seventh question finds the gap the first six leave, how the three passes separate the answer from its standing and its hole, and how to divide the work between the reader and a retrieval machine. The output of a session is not a fact. It is an experiment.

Part I. Scientific Knowledge Connectome

A connectome is the complete wiring map of a nervous system. Every cell, and every connection each cell makes. The cells alone tell you almost

nothing. The wiring is where the behavior lives.

Scientific knowledge has the same shape. The facts are the cells. The connections between them are the wiring, and the wiring is almost never drawn. Each field keeps its own facts in its own journals and its own vocabulary. Nobody owns the map between them.

The Companion is an attempt at that map. The 143,720 internal links are its wiring. The five sections below are how you work it.

This is the real descriptor of what the volume is about. Its shape, its size, its function, its process. Its dimensions, its rotations, its twist, its quaternions. Its geometry: linear, planar, bond angles.

Biological machinery. Micromachines. Enzymes, proteins, lipids, fats, carbohydrates, sugars, cofactors, vitamins, minerals.

Animal, vegetable. Plants, herbs, flowers, trees, branching. The nervous system. Blood, arteries, vessels, veins. Rivers, flow. Kidney loops and tubes, water management, pressure release.

Salts, metals, soft metals. The atomic elements and the periodic table. Physical constants and algorithms.

1. Exploration

Exploration is the second way to use the book. The first way is the query: you bring a question and the loop closes on it. Exploration has no question at the start.

You open the volume at a section and follow a link. Then another. You are not looking for an answer. You are looking at what sits next to what.

This is not idleness. A query can only find what your question already reaches toward. You cannot ask about a connection you have never imagined,

and the connections you have never imagined are the ones worth having. Exploration is how you find the question that the query will later answer.

Set a rule before you start so the session ends: follow ten links, or follow links for twenty minutes. Write down every pairing that surprises you. Most will be nothing. One or two will become questions.

2. Edge

The edge is where the literature stops.

Every field has one. It is rarely marked, because a paper reports what was found, not what was not looked for. The edge shows itself in negative space: a mechanism everyone cites and nobody measured, a figure from one experiment in 1974 that has been quoted ever since, a step that every review draws as an arrow.

The seventh question is an edge detector. When an answer comes back verified and complete but not validated, you have found an edge. The literature is consistent there and has not been tested there.

Mark every edge you find, even when it is not the edge you came for. Edges do not move often. The one you log today is still there next year, and it is where experiments come from.

3. Domain adjacent

The answer to a question in one field often sits in the field next door.

The fields do not talk. A sulfur chemist and a retinal specialist read different journals, attend different meetings, and use different words for the same molecule. A fact that is ordinary in one is unknown in the other. Nobody is hiding it. The two literatures simply never touch.

Domain adjacent is the deliberate move across that line. You have a question in field A. Before you conclude the answer does not exist, you ask which field handles the same chemistry, the same geometry, or the same failure mode under a different name, and you go look there.

The volume crosses these lines on purpose. Sulfur runs through 1,581 of its pages, and those pages sit in geology, in biochemistry, in cell biology, and in medicine. That is one atom read across four fields that do not cite each other.

When you run the octave, ask the adjacency question at *where*. Where does this take place is not only a location in the cell. It is also a location in the literature.

4. Mapping connections

A connection is a claim, and it gets recorded like one.

When you find that two things connect, write down four items: the two sections, what kind of connection it is, and what would break it. A connection with no breaking condition is an association, not a finding.

The kinds worth separating:

- **Shared mechanism.** The same chemistry does the work in both places.
- **Shared constraint.** Different mechanisms, the same limit on both.
- **Sequence.** One is upstream of the other.
- **Analogy.** The shapes match and nothing physical is shared. Useful for thinking, never evidence.

Keep analogy in its own category and label it every time. An analogy that loses its label becomes a mechanism in somebody's head within two readings.

5. Knowledge networks

The map is worth more than any node on it.

A fact answers one question. A connection answers a class of questions, because it carries you from a field where you are stuck to a field where the work is already done. That is why the wiring is the asset.

The network also shows you things no single section can. A node with many links into it is load-bearing: if it is wrong, a great deal moves. A node with one link is either a dead end or an unexplored direction, and the two look identical until somebody goes down it. A region with dense internal links and almost nothing crossing out of it is a field talking to itself.

What the Companion is building is this: a network of scientific knowledge where the connections are written down, so the next person does not have to rediscover each one by accident. The facts are already published. The map is not.

Part II. The Book

6. What this book is

The Berger Scientific Companion is a lookup table. You query it. You do not read it.

A book you read has a first page and a last page. The author sets the order. The Companion has no such order. It has 2,680 numbered sections and 143,720 internal links. Every section is an entry point. You arrive wherever your question lands, and the links carry you out from there.

It is a scientific answering machine. You bring a question. It returns the answer and the section the answer sits in. Then it shows you where the answer runs out.

That last part is the reason the book exists. Most reference work stops at retrieval. You look up a fact, you get a fact, you close the book. The Companion does not let you stop there. It requires you to name which part of your answer is unsupported before you use the answer.

7. What is in it

These are the numbers on record. Where this manual gives a figure, that figure is the fact.

item	figure
pages	3,543
numbered sections	2,680
internal links	143,720
file size	79 MB
books in the volume	2

The file is NovagonDNA_Combined_Master_v332.pdf. Book I is PP-2026-MASTER-01. Book II is PP-2026-VTI-01. The title of the volume is *The Scientific Arc of Man's Evolution from the Big Bang to the Human Genome*.

The volume is deposited at Zenodo under concept DOI 10.5281/zenodo.23827226. The repository is github.com/drjab699-jab/scientific-arc.

The book is color-coded and hyperlinked throughout. Most Parts close with the same four blocks: what this Part claims, what this Part refuses, what would show this Part is wrong, and the experiment this Part generates. The

experiments are numbered and run in their own series.

Part III. The Method

8. Bring a question, not a topic

The loop runs on questions. It stalls on topics.

“Sulfur” is a topic. It returns 1,581 pages. You cannot audit 1,581 pages.

“Why does the cell keep a radical parked on a cysteine in ribonucleotide reductase?” is a question. It has a subject, a verb, and a thing you do not yet know. The loop can close on it.

Test your question before you run it.

1. Does it name one thing? If it names three, you have three questions. Run them one at a time.
2. Does it ask something you do not already know? A question you can answer from memory returns a confirmation, and a confirmation teaches you nothing.
3. Could the answer be wrong? If no answer to it could be wrong, it is not a question. It is a definition. Look it up and move on.

If you cannot get your question down to one sentence, you have not finished thinking about it yet. Think about it. Then come back.

9. The Berger octave loop

Six questions in a ring: who, what, where, when, how, why, and beginning again at who. Each one opens into its own set.

- **Who?** Who is the person? Who is the place? Who is the thing?

- **What?** What is the purpose? What is the reason? What is the expectation?
- **Where?** Where does it take place? Here, there, someplace else.
- **When?** When does it occur? Past, present, future.
- **How?** How do you do it? What is the process? What is the sequence? What is the structure? What is the function? What is the shape?
- **Why?** Why are you doing it this way? Why did you make your process choices? What are the first principles you are trying to explain or look up?

Ask all six. Do not skip the one that looks obvious. The question you skip is usually the one holding the gap, because you skipped it on an assumption you never checked.

Write the six answers down. You need them side by side for the seventh question.

10. The seventh question

Which one of these six questions is not verified, complete, or validated?

This is the engine of the loop. The first six retrieve. The seventh finds the hole.

The three words are three different tests. Run each of them on each of your six answers.

test	what it asks	what a failure looks like
verified	Is there a source behind this, and did you open it?	The answer traces back to a review that cites a review. Nobody opened the measurement.
complete	Does the answer cover the whole question, or part of it?	The *how* covers the process and says nothing about the shape.

validated	Has anyone tested this against the world, or is it only consistent?	The mechanism is coherent on paper and no experiment has touched it.

An answer can pass one test and fail another. A number measured once in 1974 and quoted ever since is verified and not validated. A mechanism everyone agrees on with no measurement under it is complete and not verified.

Name the weak one out loud. Write it in the margin, in the file, in the note. An unmarked gap is worse than no answer, because you will build on it.

11. Run it three times

One pass is a lookup. Three passes are an audit. Each pass asks the same six questions of a different object.

Pass one asks about the subject. You put the six questions to the thing you came to find out about. The sections give you what they have. You come out with an answer.

Pass two asks about the answer. Same six questions, new object. Who established this. What was the purpose of establishing it. When was it settled, and has anything moved since. How was it measured. Why is it stated this way. You come out with the answer's standing in the literature.

Pass three asks about the gap. Now the seventh question has something to work with. You know the answer and you know where it came from, so you can see which of the six is thin. This pass produces the experiment.

Do not collapse the passes. Pass two feels redundant the first few times you run it, and it is where most of the work gets done. An answer you have not asked *who* and *when* about is an answer you have taken on trust.

12. The theory of threes and the bonding partner

Two more checks run on every answer.

The theory of threes. Past, present, future. X, Y, Z. Breadth, length, height. Nothing in the world is one-dimensional, and an answer that holds at one time or along one axis is not an answer about a thing. It is an answer about a slice. Put the third dimension on it and see whether it survives.

The bonding partner. Nothing in biology stands alone. For every answer, ask what it is bonded to, and then walk the ladder.

Atoms, molecules, cells, types, tissues, organs, glands. Chemical, biochemical, biology, molecular biology. Genetic information, genetic code. Nervous system, nerves, senses, inference. Sight, sounds, smells, tastes: sweet, sour, bitter, salty, umami.

Walk it up and walk it down. A chemical answer with no cell behind it is a loose end. A clinical answer with no atom under it is a loose end. An answer that exists at one rung and nowhere else is not finished, and the rung where it breaks is where your next question lives.

The sulfur thread through the volume is this ladder worked all the way out. One atom, tracked from its electron shell to a disease, with no rung skipped in between.

Part IV. The Session

13. Using the Companion with a machine

No person holds 2,680 sections in working memory. No specialist in any one of the fields this volume crosses holds them either. The book needs a retrieval organ, and that is what the machine is for.

The division of labor is fixed.

you supply	the machine supplies
the question	the sections that bear on it
the judgment about what matters	the section numbers and the text
the decision about what is load-bearing	a plain report of which of the six questions came back unsupported
the next experiment	the check against the published literature

The machine finds and returns. It does not decide what is important. That rule is not politeness. A retrieval system that ranks findings by its own sense of significance hands you the consensus every time, and the consensus is exactly what the seventh question is built to interrogate.

How to run a session. Give the machine the question in one sentence. Tell it which pass you are on. Have it pull sections by number and quote them back with the numbers attached. At the end of each pass, ask it directly which of the six came back unsupported, and have it say so plainly.

What to refuse from the machine. An answer with no section number. A summary that smooths over a disagreement between two sections. A confident statement where the volume is silent. Silence in the volume is a finding, not a failure, and the machine reports it as silence.

14. A worked example

The question. Why does ribonucleotide reductase park a radical on a cysteine?

Pass one, the subject. The loop goes to §4, Sulfur and the Electron. Who: the enzyme, and the sulfur atom holding the unpaired electron. What: the purpose is to pull the 2′ hydroxyl off a ribonucleotide and make a deoxyribonucleotide. Where: in the cell, at the step where RNA chemistry becomes DNA chemistry. When: before DNA existed as a storage molecule, and in every dividing cell since. How: the radical sits on sulfur because sulfur's outer electrons are loosely held and the atom tolerates an unpaired electron that oxygen does not. Why: the cell needs a radical it can park and retrieve, not one that destroys whatever it touches.

The answer: sulfur's polarizability makes it a stable seat for a radical.

Pass two, the answer. Who established the polarizability figure, and when. Was it measured in the enzyme or in a model compound. Has anything moved since. Here the loop finds that the chemistry is old and settled, and that the measurements come from model systems rather than from the enzyme at work.

Pass three, the gap. Run the seventh question across the six. *How* is verified and *how* is complete. *How* is not validated in the enzyme itself. That is the thin one.

The output. Not a fact. An experiment: measure the radical's seat in the working enzyme, not in the model compound. The Parts of the volume end on exactly this kind of statement, and they number them. Experiment Sixty-Three sits at §1400.

Notice what the session produced. It started as a lookup and ended as a piece of work somebody can do at a bench. A session that produces only confirmations has not used the book.

15. House rules

1. **Query it. Do not read it.** Bring a question, not a topic.
2. **Cite by section, not page.** Sections are stable. Pages drift. In Book I the PDF page runs about five ahead of the printed folio, so PDF 1423 is printed I-1418. Write §4, not p.103.
3. **The numbers on record are the facts.** Where the volume gives a figure, use that figure. Do not round it, soften it, or replace it with one you remember.
4. **Run the octave three times.** Answer, context, gap. No shortcuts through pass two.
5. **Walk the scale ladder.** An answer that lives at one rung is not done.
6. **End on the gap.** Name which of the six came back unsupported. That is the output of the session.
7. **Silence is a finding.** If the volume does not treat something, say so and mark it. An absence is a place to work.

On searching the file: run pdftotext with the layout flag on the PDF and grep the result. The whole book never has to be loaded. Section headings carry a number, a period and the heading text.

Part V. The Origin

16. Where the book came from

John Allen Berger started with over 4,000 scientific medical books bought throughout every Big Ten university in the late 1990s and early 2000s. He went from campus to campus. He looked at every table of contents and every index at the end of the book. He browsed through the color-coded, colorful pictures, diagrams, and figures.

Over three years he took over 100,000 JPEGs. He then began his project, finally putting about 600 of them on his wall in an 8-foot by 9-foot mural. That mural is described in the 3,543-page book. The seventy-two-panel wall and the sentence “The thread is sulfur” are at page 90.

The method came from storyboarding and screenplay writing. A screenplay is built by asking who, what, where, when, how and why of every scene until the scene holds together or shows you it does not. Berger put the same questions to the literature. The Companion is the result, and the octave loop is the instrument.

The work is released openly. Berger wants the next wave of young scientists to have something new to explore. The book is not a monument to a finished argument. It is a working surface, built so somebody who was not there for any of it can come in cold, ask a question, and be handed both the answer and the place where the answer runs out. The second half of that is the gift.

Status. This manual accompanies NovagonDNA_Combined_Master_v332.pdf, 3,543 pages, 2,680 numbered sections, 143,720 internal links, deposited at Zenodo under concept DOI 10.5281/zenodo.23827226 and mirrored at github.com/drjab699-jab/scientific-arc. Open for distribution under CC BY 4.0.